Jimin Hong

Ph.D. Student · Vehicle-Integrated Photovoltaics & Energy Systems

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RESEARCH INTERESTS

Predicting and improving the power-generation performance of vehicle-integrated photovoltaics (VIPV) in real driving and parking environments: 360° sky imaging and solar access analysis, solar EV parking optimization, AI-based multi-surface generation forecasting, off-grid renewable microgrids and energy storage, and CCUS-integrated energy system optimization.

EDUCATION

Integrated M.S.–Ph.D. Program, Energy Resources & Geosystems Engineering — Sep 2023 – Present
Sejong University, Seoul, South Korea · Advisor: Prof. Yosoon Choi

B.S., Energy Resources Engineering — Mar 2020 – Aug 2023
Pukyong National University, Busan, South Korea

JOURNAL PUBLICATIONS

1. **Hong, J.**, & Choi, Y. (2025). Advanced parking assistance system for solar electric vehicles using 360° virtual reality imaging and real-time solar radiation forecasting. *Applied Energy*, 397, 126365. [SCIE]
   doi.org/10.1016/j.apenergy.2025.126365
2. **Hong, J.**, & Choi, Y. (2025). Design and implementation of a solar access database for optimizing parking of solar electric vehicles. *Solar Energy Materials and Solar Cells*, 293, 113841. [SCIE]
   doi.org/10.1016/j.solmat.2025.113841
3. **Hong, J.***, Khavari, F.*, & Choi, Y. (2026). Optimizing CCUS–methanol integrated systems using a two-stage framework: Enhancing profitability and sustainability of coal in the carbon-neutral era. *Korean Journal of Chemical Engineering*. (*co-first authors) [SCIE]
   doi.org/10.1007/s11814-026-00742-y
4. **Hong, J.**, & Choi, Y. (2026). Comparative techno-economic assessment of energy storage technologies in an off-grid renewable microgrid for AI data centers. *Journal of Engineering Research*. In press.
   doi.org/10.1016/j.jer.2026.10.038
5. Choi, S., **Hong, J.**, & Choi, Y. (2025). Development of Arduino-based power-generation data logger for evaluating potential of vehicle-integrated photovoltaic systems. *Journal of the Korean Society of Mineral and Energy Resources Engineers*, 62(4), 414–424.
   doi.org/10.32390/ksmer.2025.62.4.414
6. Choi, Y., Suh, J., Kim, S.-M., Ku, J., **Hong, J.**, & Park, H.-D. (2025). Research trends on solar electric vehicles in Korea. *Journal of the Korean Society of Mineral and Energy Resources Engineers*, 62(1), 41–60.
   doi.org/10.32390/ksmer.2025.62.1.041
7. **Hong, J.**, Baek, J., & Choi, Y. (2023). Development of parking space analysis system for solar electric vehicle using 360 VR camera. *Journal of the Korean Solar Energy Society*, 43(3), 11–30.
   doi.org/10.7836/kses.2023.43.3.011

AWARDS & HONORS

2026 Best Paper Presentation Award, 2026 Fall Joint Conference of the Korean Resources Societies

2026 Silver Prize (KOMIA President's Award), 7th Mine-Tech Festa, Innovative Technology Competition

2026 Best Paper Presentation Award, 126th Spring Conference, Korean Society of Mineral and Energy Resources Engineers

2025 Encouragement Award, 6th Mine-Tech Festa, Korea Mine Rehabilitation and Mineral Resources Corp.

2024 Best Paper Presentation Award, 2024 Fall Joint Conference of the Korean Resources Societies

2024 Presidential Commendation, Korean Association of Geographic Information Studies

2023 Special Prize (Renewable Energy), 1st Energy Optimization Design Contest, Korean Solar Energy Society

2023 University President's Award, Pukyong National University

2023 Outstanding Paper Award (Korea Energy Agency), 2023 Spring Conference, Korean Solar Energy Society

2023 Outstanding Oral Presentation Award, Korean Solar Energy Society

2022 Excellence Award, 2022 Energy Data & AI Idea Contest, Korean Solar Energy Society

CONFERENCE PRESENTATIONS

2026 2026 Fall Joint Conference of the Korean Resources Societies — AI agent-based prediction and automatic correction of five-surface power generation for parked solar EVs (Best Paper Presentation Award)

2026 PV in Motion 2026 [International]

2026 126th Spring Conference, KSMER — Energy storage systems for an off-grid RE100 microgrid for AI data centers (Best Paper Presentation Award)

2026 Spring Conference, Korean Society for New and Renewable Energy

2026 SME Annual Conference & Expo 2026 [International]

2025 AFORE 2025, Asia-Pacific Forum on Renewable Energy [International]

2025 Fall Conference, KSMER

2025 PV in Motion 2025 [International]

2025 Spring Conference, Korean Solar Energy Society

2025 Spring Conference, Korean Society for New and Renewable Energy

2024 APSRC 2024, Asia-Pacific Solar Research Conference [International]

2024 2024 Fall Joint Conference of the Korean Resources Societies — Solar Access database for solar EV parking (Best Paper Presentation Award)

2024 PV in Motion 2024 [International]

2024 Spring Conference, Korean Association of Geographic Information Studies

2024 Spring Conference, Korean Solar Energy Society

2023 Fall Conference, KSMER

2023 2023 Spring Conference, Korean Solar Energy Society — Parking space analysis system for SEVs using a 360 VR camera (Outstanding Paper & Oral Presentation Awards)